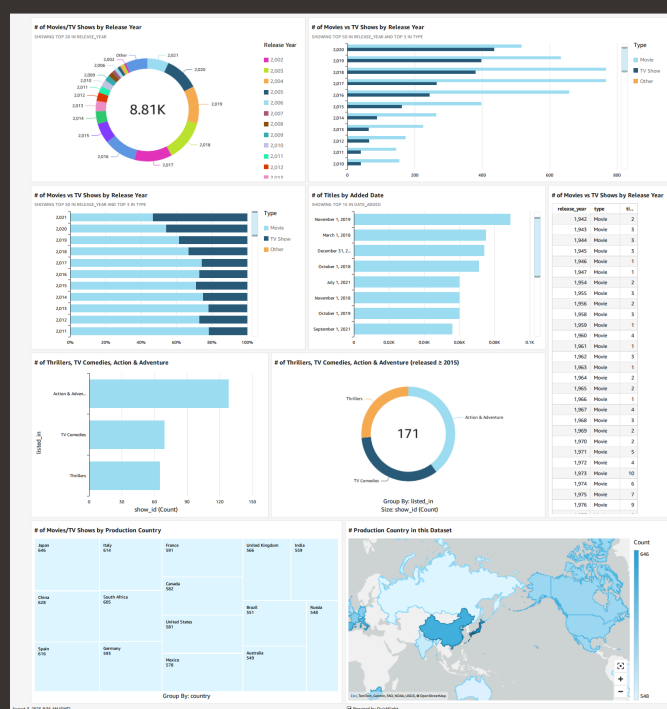




Muhammad Abbas





Introducing Today's Project!

In this project, I will demonstrate how to use Amazon QuickSight to analyse Netflix data and generate visualizations and insights! I'm doing this project to learn how to use cloud data services for data analysis.

Tools and concepts

Services I used were QuickSight and S3. Key concepts I learnt include manifest.json, data visualization techniques (e.g. charts, filters) and how to perform a data refresh.

Project reflection

This project took me approximately 2 hours. The most challenging part was understanding how a manifest.json file works. It was most rewarding to generating a PDF of my finished visualization.

After this project, I plan to work on more visualization projects. I will start this project as soon as I can ;)

Upload project files into S3

S3 is used in this project to store two files, which are manifest.json (which tells QuickSight about the structure and format of the data we're analyzing) and netflix_titles.csv (which is the raw data that we're here to analyse today)

I edited the manifest.json file by updating the S3 URI that corresponds to our dataset's file location. It's important to edit this file because it's how QuickSight will find and analyse the data if we didn't update this file, QuickSight wouldn't know the correct location of my dataset. This will casue errors down the line.

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

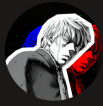
Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

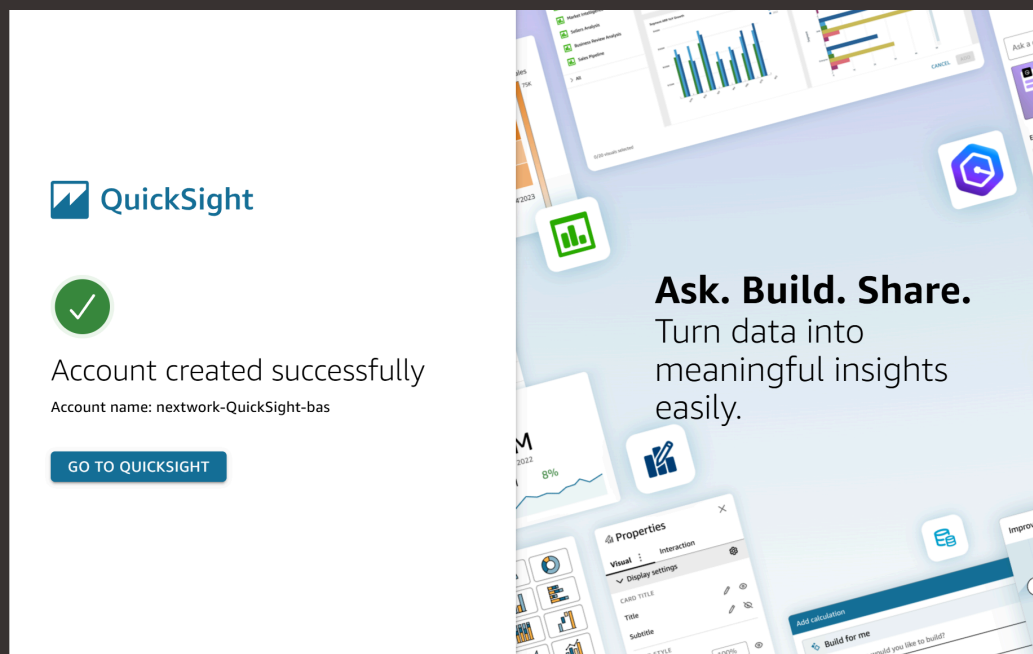
	Name	Type	Last modified	Size	Storage class
☐	manifest.json	json	August 1, 2025, 04:21:15 (UTC+08:00)	307.0 B	Standard
☐	netflix_titles.csv	csv	August 1, 2025, 04:19:55 (UTC+08:00)	3.2 MB	Standard



Create QuickSight account

Creating a QuickSight account cost \$0 as comes with a 30 day free trail!!! Make sure you UNCHECK an add-on in the sign up flow called Pixel-Perfect Reports so you don't get charged.

Creating an account took me around 10 minutes, including setting up my S3 bucket.





Download the Dataset

I connected the S3 bucket to QuickSight by visiting Datasets page. There were SO many options for data sources we could connect to (databases and external tools/platforms like Salesforce), and I selected S3.

The manifest.json file was important in this step because it tells QuickSight how to read the data - in this project, it tells QuickSight that we're uploading a .CSV file (spreadsheet) and the delimiter (i.e. commas) so that the QuickSight knows how to break up the data for analysis. Otherwise, QuickSight might get confused.

New Amazon S3 data source

Data source name

kaggle-netflix-data

Upload a [manifest file](#)

☒ URL ☐ Upload

s3://nextwork-quicksight-project-bas/manifest.json

Connect

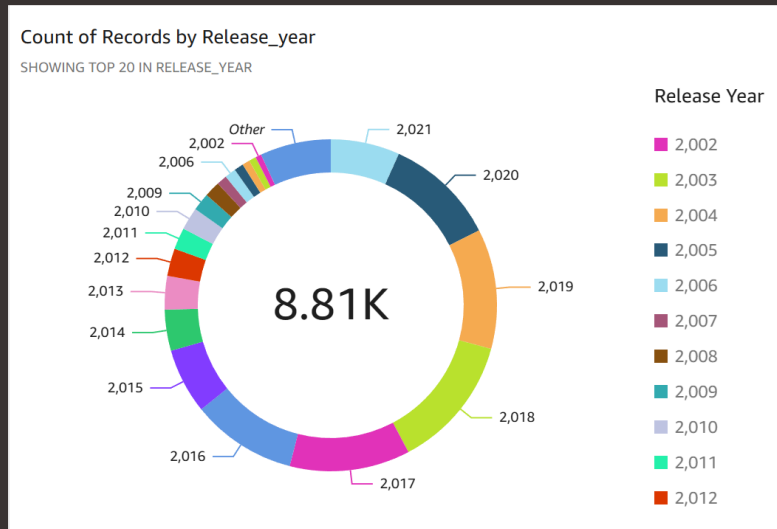


My first visualization

To create visualizations on QuickSight, I simply have to click on data field/labels e.g. `release_year` and QuickSight will automatically generate a graphic that best suits that type of data. You can also drag data labels into sections like "Group By" or "Y-Axis" to determine how the graphic should treat the data.

The chart/graph shown here is a breakdown of the release years of the content inside Netflix - i.e. how many TV shows/movies were released on xyz year? You can see the there is a total of 8800+ content pieces, and 2019 seem to be the most amount of content released.

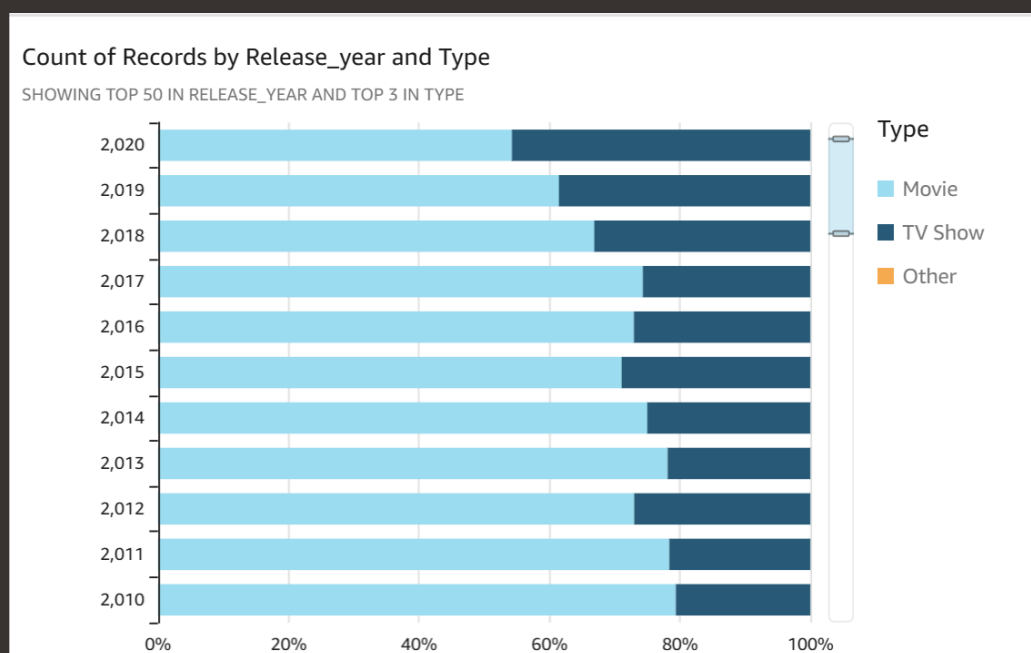
I created this graph by simply clicking on the '`release_year`' data label and changing the automatically generated chart from a bar chart to a donut chart.

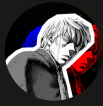


Using filters

Filters are useful for narrowing down our data to the subset that we want to focus on, and in this case., we could use filtes to focus on the specific categories that we want to analyze! We also used filters to only look at content with a released date from 2015 and beyond.

This visualization is a breakdown of. TV shows/movies that belong in one of three categories - action and adventure, tv comedies and thrillers. Here I added a filter based on the "listed on" data label i.e. only these three categories could pass the filter.

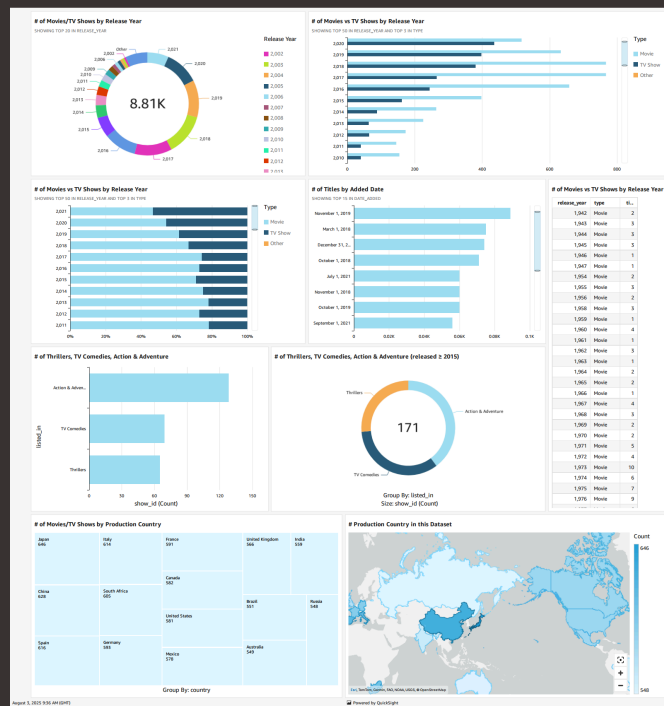




Setting up a dashboard

As a finishing touch, I updated the titles of the charts in my dashboard so that they are really readable. The default names would simply mention the data label e.g. released_year but the now titles communicate the purpose of a chart clearly e.g. # of Movies vs TV Shows by Release Year.

Did you know you could export your dashboard as PDFs too? I did this by selecting Publish and then 'Generate PDF' on the top right hand corner of the QuickQuick analysis.



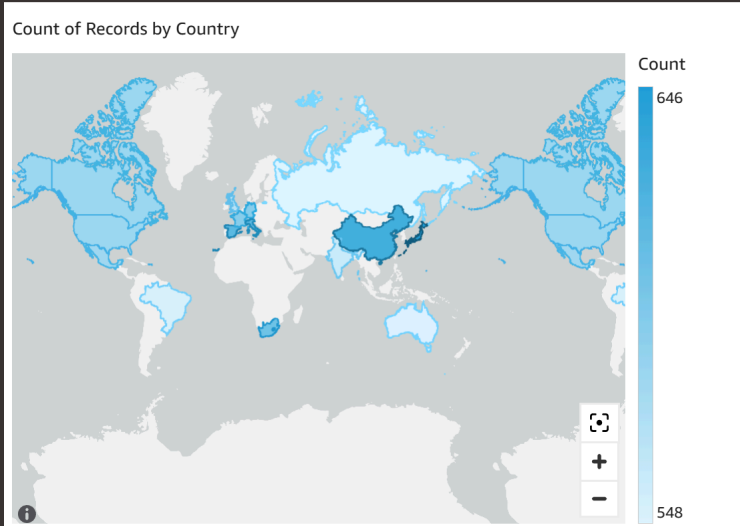


Refreshing source data

In this project's extension, I downloaded fresh data that's different from my original dataset because it had rows and rows of empty Country data. Analysing incomplete data brings the risk of generating inaccurate insights, which leads to the wrong business decisions that cost the company time, effort and money.

Once I downloaded new data, I had to update my S3 bucket because it is still storing the previous version of the data (i.e. the one with country data missing). I also uploaded a new manifest.json file that points to the updated dataset's file name. This makes sure that QuickSight is now pulling in data from the updated dataset and not the version with missing data.

I initially couldn't see my updated data in QuickSight, so I had to visit the dataset in the Datasets page in QuickSight, and perform a full refresh of the data!





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